Harsh Thomare

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EDUCATION

University of California, Irvine **December, 2024**
B.S. in Computer Science and Engineering *Irvine, CA*

- **Honors:** UROP Distinguished Research Fellowship, Sigma Prize Winner in Innovation Challenge

WORK EXPERIENCE

Mainspring Energy | Embedded Software Engineer | Menlo Park, CA | *07/2024 – Present*

- Engineered a sensorless back-EMF tracking algorithm for 60kW linear motors on **bare-metal** STM32F4, increasing stopping force by 41% and reducing thermal accumulation by 30%.
- Designed and implemented a C-based **UART** bootloader for Docker-based **OTA** updates, incorporating A/B bank switching and CRC32 verification for 1,000+ distributed devices.
- Optimized **FOC** for 3-4kW PMSM motors, achieving a 3% efficiency gain and reducing peak temperatures by 15°C.
- Developed a Python **Software-in-the-Loop** (SIL) framework to validate **C++ firmware** logic against a physics model, reducing **hardware-in-the-loop** debugging time by 3+ days.

Mainspring Energy | Embedded Software Engineering Intern | Menlo Park, CA | 06/2023 – 12/2023

- Migrated STM32 firmware to **FreeRTOS**, saving 20% memory by consolidating **LwIP** message queues and optimizing task stack allocation.
- Developed a distributed, low-latency communicator using a **Pub-Sub** architecture with **DDS** middleware to synchronize telemetry across **Ethernet**-linked controller systems.

Second Order Effects | Electrical Engineering Intern | El Segundo, CA | 01/2024 – 04/2024

- Engineered a high-speed data capture engine for 64+ ADCs via **SPI** on an Xilinx FPGA, utilizing Verilog and on-chip memory for 100MSPS sustained throughput.
- Developed **SystemVerilog/C++ UUT** drivers with hardware timestamping to validate data integrity across high-speed acquisition cycles.

RESEARCH EXPERIENCE

AI Cyber-Physical Systems Lab | Research Assistant | Irvine, CA | 05/2025 – 11/2025

- Co-authored **HELIOS** (arXiv:2601.14598); built a Ghidra-based pipeline to extract structural representations for **LLM-enhanced decompilation**.
- Developed a structurally-grounded LLM for **binary-to-C translation**, achieving a 5.8% improvement in re-compilability and re-executability over SOTA baselines.

PROJECTS

IEEE Micromouse Competition | 09/2021 – 06/2023

- Designed a multi-layer STM32F2 schematic in **KiCAD** and debugged via **oscilloscope** and **logic analyzer**.
- Programmed C-based APIs for motors and encoders; implemented **PID** control and **A*** algorithms for autonomous path planning.

IEEE@UCI Macropad Workshop | Project Lead | 02/2023 – 05/2023

- Led hardware workshop for 60+ students, teaching PCB design and QMK firmware programming; secured $1,600 in funding.